INCH-POUND

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2 January 2009
SUPERSEDING
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DETAIL SPECIFICATION

CANDLES, ILLUMINATING, SURVIVAL, HIGH MELTING POINT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high melting point candles of 10-hour duration.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Standards and handbooks. The following standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil/.

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DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-831 - Preparation of Test Reports

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article testing in accordance with 4.3.

3.2 Materials.

3.2.1 Raw materials. The manufacturer is permitted wide latitude in the selection of raw materials provided the candles meet the requirements of this specification.

3.2.2 Wick. The wick of the candle shall be of a satisfactory quality braided yarn.

3.3 Performance.

3.3.1 Effect of temperature. The candle, when tested as specified in 4.5.2, shall not show that wax flowed after a period of not less than 24 hours, at +125 °F ± 3 °F.

3.3.2 Deformation. The candle shall not deform when tested as specified in 4.5.2.

3.3.3 Burning quality. The flame shall consume the candle in not less than 10 hours when burned as specified in 4.5.3, without dripping, excessive smoking, or objectionable odor, and without the necessity of trimming the wick.

3.4 Weight. The weight of the candle, including the can specified in 3.5, shall be not more than 130 grams.

3.5 Size. The candle shall be contained in a .012 aluminum round slip cover can, nominal size 2.75 inches diameter by 1.75 inches high, body inside lock seam, top edge of body plain, single external bead near top of body for plain edge cover to seat against, bottom flanged, doped double seam with side seam striped with vinyl lacquer. Fill the container with wax to 1.25 inches. Seamless can with same nominal dimensions will be acceptable.

3.6 Sealing. The can lid shall be compression fit and shall not come off unintentionally.

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3.7 Workmanship. The product shall be prepared by such methods as will reflect good standards of workmanship and quality in the finished product. The wick of the candle shall be properly centered.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.2).

b. First article testing (see 4.3).

4.2 Conformance inspection. Conformance inspection shall consist of sampling plans and tests.

4.2.1 Sampling plans and tests.

4.2.1.1 Sampling plan A. Candles shall be inspected by sampling methods in accordance with MIL-STD-1916, for the examination of product test, as described under 4.5.1.

4.2.1.1.1 At the discretion of the inspecting agency, any action specified in MIL-STD-1916 for accomplishment by the Government shall be interpreted to mean action to be accomplished either by or under the supervision of the Government, with the final decision as to acceptance or rejection in either case resting with the Government inspector.

4.2.1.2 Sampling plan B.

4.2.1.2.1 Lot. For purpose of the sampling plan test, a lot shall consist of all candles which are constructed from the same batch of materials, and submitted for inspection at substantially the same, time.

4.2.1.2.2 Sampling tests. Eight candles shall be selected at random from each lot and subjected to the tests listed below as described under 4.5.

a. Effect of temperature (see 4.5.2).
b. Burning quality (4.5.3).

4.2.1.3 Rejection. When one or more items fail to meet the specified sampling plan tests, acceptance of all items in the lot shall be withheld until the extent and cause of failure are determined. The lot for a repeat of the sampling plan tests, the contractor shall fully explain to the inspector both the corrections made and the cause of previous rejections. Before resubmitting the lot for a repeat of the sampling plan tests, the contractor shall fully explain to the inspector both the corrections made and the cause of the previous rejections.

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4.3 First article testing.

4.3.1 First article test sample tested by the contractor. The contractor shall subject candles to the first article tests specified in 4.3.4.

4.3.2 First article test report. After the contractor completes the first article tests, he shall prepare a first article test report and furnish a copy of the report to the government procuring activity. The contractor may refer to the guidelines outlined in MIL-HDBK-831 for preparing the report.

4.3.3 First article test sample. The contractor shall submit one sample to the government procuring activity.

4.3.4 First article tests. First article tests shall consist of all tests described under 4.5.

4.4 Test conditions.

4.4.1 Atmospheric conditions. Unless otherwise specified (see 6.2), all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of +77 °F, ± 18 °F, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.5 Test methods.

4.5.1 Examination of product. Sample candles shall be examined to determine compliance with the workmanship, weight, and dimensions specified herein.

4.5.2 Effect of temperature. The candle shall be placed with axis horizontal (can lying on side) in a constant temperature cabinet and maintained at +125 °F, ± 3 °F, for not less than 24 hours and the specimen observed periodically. The candle shall not show deformation. Wax flow shall be cause for rejection.

4.5.3 Burning quality. The flame shall consume the candle in not less than 10 hours under conditions free from drafts and at an ambient temperature of plus +70 °F, ± 5 °F, without dripping, excessive smoking, or objectionable odor and without the necessity of trimming the wick.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the

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managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The candles covered by this specification are intended for use in under emergency survival conditions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article testing is required (see 3.1).
c. Atmospheric conditions, if different (see 4.4.1).
d. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

10-hour duration
Aluminum can
Wick

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL
Air Force - 99
DLA - IS

Preparing Activity:
DLA - GS2

(Project 6260-2009-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.daps.dla.mil/.

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